Venkata Lakshmi Parimala Pasupuleti

AI Engineer · Agentic AI Systems · RAG Pipelines · LLM Evaluation · Python · AWS · Production AI · Regulated Environments

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SUMMARY

AI Engineer with 9+ years of production engineering depth and two years building agentic AI systems at scale. Built **Leasel**, a production 8-agent LangGraph system that plans, reasons, and acts across complex compliance workflows - durable state, tool use, conditional routing, HITL escalation, and audit logging throughout: the same intelligent adaptive system architecture American Express is building for financial services. Built **AgentProbe**, an open-source LLM evaluation toolkit with a CI-integrated regression harness, golden datasets, and LLM-as-judge scoring - evaluation and monitoring tooling that keeps production AI quality from degrading over time. At Entrata (9+ years, 100K+ users), owned mission-critical systems in a regulated environment requiring the same auditability, explainability, and zero-tolerance-for-errors standards as financial services: PHI-encrypted data pipelines, HUD/LIHTC compliance workflows, and 7+ years of zero violations. Strong Python, AWS, LangGraph/LangChain, RAG pipelines, event-driven systems (RabbitMQ/Kafka-pattern), REST APIs, and AI-assisted development. M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

CORE COMPETENCIES

Agentic AI Systems · LLM APIs · Prompt Engineering · RAG Pipelines · Hybrid Search · LLM Evaluation · Monitoring Tooling · LangGraph · LangChain · Python · FastAPI · Django · Schema Validation · Structured Outputs · Vector Storage · Embeddings · AWS (S3 · EC2 · RDS · Lambda) · GCP · Azure OpenAI · Kubernetes · Docker · Event-Driven Architecture · Kafka · REST APIs · gRPC · CI/CD · GitHub Actions · HITL · Guardrails · Audit Logging · Regulated & Compliant Systems · Observability · Tracing · AI-Assisted Development

EXPERIENCE

AI/ML Engineer (Volunteer)

Sep 2026 - Present

Blue Horse Digital — Virginia, USA

- Building ML models, analytical data pipelines, and AI-powered application features; connecting AI components to backend APIs and reporting layers; debugging data-driven features across the stack.

AI/ML Engineer Intern

May 2025 - Aug 2025

Magizhs Tech Solution

- Built a production agentic workflow (LangGraph StateGraph) with RAG over regulatory documents, schema validation guardrails, and conditional routing - natural-language inputs to structured, validated outputs; deployed MCP server connecting agentic pipeline outputs to downstream enterprise systems.
- Engineered full document ingestion, chunking, retrieval, and FastAPI streaming pipeline with defined API contracts and retry semantics; operated and monitored the system post-launch.

Senior Software Engineer

Sep 2018 - Aug 2024

Entrata - PropTech / Real Estate SaaS (100K+ users, regulated multi-tenant)

- **Owned mission-critical compliance systems in a regulated environment** for 7+ years - HUD income-limit data pipeline (few thousand federal rows → ~180,000 domain records), eligibility/rent/HAP calculations for 100K+ housing units; PHI-encrypted AWS S3 pipelines (Libsodium); 7+ years of zero compliance violations and zero security incidents - the same auditability and reliability standards required for production AI in financial services.
- **Built and operated event-driven async processing** (RabbitMQ) for tenant record updates - retry logic, idempotency guarantees, and production incident ownership across the full distributed system lifecycle; monitored, debugged, and improved reliability post-launch.
- **Drove observability and reliability improvements** - built internal debugging tools that reduced debugging time by 40%; led postmortems and root-cause analysis across services; maintained bug count below 10 for 6+ consecutive months across a platform serving 100K+ users.
- Optimized 200+ SQL queries (30% gain) across complex multi-tenant PostgreSQL schemas; shipped 20+ full-stack production modules (Python, PHP, JavaScript) across 120+ Agile sprints; led microservices redesign (Docker) - 50% scalability improvement, 30% deployment time reduction.
- Mentored 5+ engineers; conducted 300+ peer code reviews; eliminated 80% of manual operational workload via Python/Bash automation pipelines.

Associate Software Engineer

Jul 2017 - Sep 2018

Entrata India, Pune

- Built property management and compliance modules (Python, PHP, Java, PostgreSQL); led AWS S3 cloud migration from legacy storage; delivered large-scale data import workflows (1.6M records) under senior engineer guidance.

PROJECTS

AgentProbe - Open-Source LLM Evaluation & Monitoring Toolkit

2025 | Python, FastAPI, LangGraph, CI/CD | github.com/venkatalakshmiparimala/AgentProbe

- **CI-integrated LLM evaluation infrastructure** - 15-endpoint FastAPI backend; 5-metric eval engine (accuracy, hallucination rate, retrieval quality, completeness, compliance); LLM-as-judge scoring with golden datasets; multi-provider routing (Anthropic, OpenAI, Azure OpenAI, Groq, Ollama); regression harness that catches agent degradations before they reach production. Built as the shared evaluation layer for Leasel and used daily as an AI-assisted development tool.

Leasel - Production Agentic AI System with RAG & HITL

2026 | Python, LangGraph (8-agent), FastAPI, SQLite, CI/CD | github.com/venkatalakshmiparimala/Lease_complaince_rag

- **8-agent LangGraph system** that plans, reasons, and acts across complex compliance workflows: durable state, tool use, conditional routing, trace-level observability, structured audit logging, and HITL escalation when confidence is insufficient - responsible AI patterns enforced in code throughout. RAG

layer: OCR-fallback ingestion, hybrid search (TF-IDF + semantic reranking), citation grounding to exact source clauses.

- CI regression gate with AgentProbe LLM-as-judge scoring; **93.6% precision and 87.9% F1** across 1,240 test cases (10 compliance rules, 5 US states + federal). Core design rule enforced in code: LLM reasons, deterministic or human logic makes every regulated compliance call.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia I GPA 3.87 May 2026

B.Tech Computer Science & Engineering — JNTU Anantapur 2017